



**Barium
crystals**



**Chunk of grey
selenium**



**Magnesium
crystals**



**Osmium
pellet**

for each element, grouping them into similar types. With this information, we can use the elements to make many things we need: a fluorine compound in toothpastes toughens our teeth and silicon crystals engineered into microchips operate our gadgets and phones.

Every element has its own story of where it comes from, what it can do, and how we use it. Let's begin a tour of every element one by one. It's going to be a fascinating journey.

Tom Jackson

Throughout this book you will find boxes with the following symbols. This is what each of them mean.



This shows the structure of an atom of an element, with the nucleus (made of protons and neutrons) at the centre and electrons surrounding it in their shells.



Electron



Proton



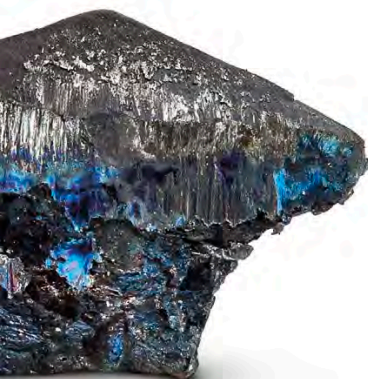
Neutron

State

The state of the element at a temperature of 20°C (28°F). It can be a liquid, solid, or gas.

Discovery

This details the year in which the element was discovered.



**Chunk of
uranium**



**Gold
crystals**



**Thulium
crystals**



**Calcium
crystals**